AUTOFIX.AI TECHNICAL WORKFLOW SPECIFICATION

End-to-End Incident Processing, Safety Arbitration, and State Transitions | Engineering Systems Manual

1. The 6-Stage Incident Lifecycle

Every incident ticket processed by Autofix-AI flows through six deterministic architectural stages:

Stage 1: Multimodal Ingestion

Ticket text and optional screenshots arrive via POST /api/tickets. The vision engine (Groq Qwen-2.5-VL / Llama-3.2) extracts raw OCR strings, hex stack traces, and UI error dialogs in < 800ms.

Stage 2: Canonical Normalization

The Error Signature Synthesizer applies regular expressions to strip ephemeral hostnames, IP addresses, usernames, and timestamps into canonical placeholders, creating a standardized diagnostic key.

Stage 3: Hindsight Memory Recall

The system queries the Vectorize Hindsight memory bank (autofix_ai_tickets). Hindsight calculates cosine similarity between the incoming key and historical resolution vectors, returning top matches and confidence C.

Stage 4: Confidence Gate Arbitration

The orchestrator evaluates C against the deterministic safety threshold (C_threshold = 0.85). If C >= 0.85, the ticket qualifies for autonomous execution. If C < 0.85, it is routed to human triage.

Stage 5A: Autonomous Remediation (C >= 0.85)

The agent executes the pre-verified PowerShell/Bash runbook in a sandboxed runner, probes system health to verify recovery, and marks the ticket RESOLVED_AUTONOMOUSLY in < 2 seconds.

Stage 5B: Human-in-the-Loop Triage (C < 0.85)

The ticket pauses in PENDING_HUMAN_REVIEW. A human engineer reviews the agent's hypothesis on the triage dashboard, approves or edits the script with one click, and executes resolution.

Stage 6: Closed-Loop Retention Hook

The verified resolution is intercepted and transmitted to Hindsight via POST /api/tickets/{id}/feedback. Long-term memory is updated, guaranteeing future recurring incidents resolve autonomously.

2. Ticket State Machine Transitions

Tickets transition through deterministic states enforced by Pydantic schemas and FastAPI handlers:

Ticket State	Trigger Event	Confidence Level	Authorized Actor
PENDING_HUMAN_REVIEW	Cold Start or Low Confidence	C < 0.85	Triage Engineer
IN_PROGRESS	Engineer opens or modifies ticket	Any	Triage Engineer
RESOLVED_AUTONOMOUSLY	High-confidence match + Health probe OK	C >= 0.85	Autofix-AI Agent

RESOLVED_HUMAN_VERIFIED	Engineer approval or runbook execution	C = 1.0 (Human)	Senior Engineer

3. Core REST API Interface Reference

- POST /api/tickets: Ingests new incident with multipart image upload and triggers vision OCR.
- GET /api/tickets/{id}: Retrieves diagnostic telemetry, OCR traces, and step-by-step reasoning.
- POST /api/tickets/{id}/feedback: HITL execution endpoint that verifies runbook and fires Hindsight retention hook.
- GET /api/memory/tree: Returns structured 5-domain knowledge tree of all learned operational runbooks.
- POST /api/memory/toggle-cloud-mode: Toggles between live Hindsight Cloud querying and Credit-Saver local cache.
- DELETE /api/tickets/{id}: Purges ticket from active storage and unlearns its vector from Hindsight.